Does $\sum_{n=1}^{\infty} \frac{7}{n^{5/2}}$ converge or diverge by the integral test?

Integral test:

If $f(n)$ is a positive, decreasing, and continuous function, then,

$\sum_{n=1}^{\infty} a_n$ diverges if

$\int_1^{\infty} a_n \, dn$

$\int_1^{\infty} a_x \, dx$ diverges.

Let $a_n = f(x)$

$$\int_1^{\infty} \frac{7}{x^{5/2}} dx$$

$$\lim_{b\to\infty} \int_1^b \frac{7}{x^{5/2}} dx$$

$$7 \lim_{b\to\infty} \int_1^b x^{-5/2} dx$$

$$7 \lim_{b\to\infty} \left[\frac{x^{-5/2+1}}{-5/2+1}\right]_1^b$$

$$7 \lim_{b\to\infty} \left[\frac{2x^{-3/2}}{-3}\right]_1^b$$

$$7 \lim_{b\to\infty} \left[\frac{2}{-3}\left(\frac{1}{x^{3/2}}\right)\right]_1^b$$

$$7 \lim_{b\to\infty} \left[\frac{2}{-3}\left(\frac{1}{b^{3/2}}\right) + \frac{2}{3}\left(\frac{1}{1^{3/2}}\right)\right]$$

Tends to 0. why?

$\frac{1}{\text{very large \#}} \Rightarrow 0$

$$7\left(0 + \frac{2}{3}\right)$$

$$\frac{7(2)}{3} = \boxed{\frac{14}{3}}$$

By the integral test,

$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{7}{n^{5/2}}$ converges,

because

$\int_1^{\infty} \frac{7}{x^{5/2}} dx$ converges,

as well.

(But, not necessarily to $\frac{14}{3}$).

* The integral test is a test for convergence / divergence.

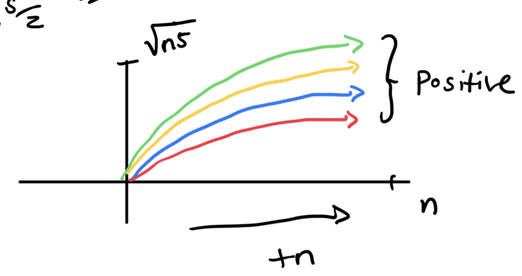
It does not give you $\lim_{n\to\infty} S_n$. (The sum of the sequence).

$[1, \infty)$

Prove:

$f(n)$ is positive

$\frac{7}{n^{5/2}} \Rightarrow \sqrt{n^5}$



then, $\sqrt{n^5}$ is positive, and therefore $f(n)$ is positive for $n \geq 1$

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Prove:

$f(n)$ is decreasing for $n \geq 1$

$$f(n) = \frac{7}{n^{5/2}}$$

$$= 7\left[n^{-5/2}\right]'$$

$$= 7\left[\frac{-5}{2} n^{-7/2}\right]$$

$$= \frac{-35}{2n^{7/2}} < 0$$

$\therefore$ $f(n)$ is decreasing on $[1, \infty)$ because $f'(n) < 0$.

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Prove:

$f(n)$ is continuous for $n \geq 1$

$$f(n) = \frac{7}{n^{5/2}}$$

$$n^{5/2} = 0$$

When $n = 0$, $f(n)$ is discontinuous.

Wait!

$n=0$ is on $[1, \infty)$,

The integral test is still valid!! :)

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